

Course Code	21CSC201J	Course Name	DATA STRUCTURES AND ALGORITHMS		Course Category	C	PROFESSIONAL CORE				L	T	P	C
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Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards		Nil	

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	know about searching and sorting techniques used to handle a set of data along with time and space complexity
CLR-2:	utilize various categories of list structures to develop solutions
CLR-3:	explore usage of Stack and Queues in processing data for real time applications
CLR-4:	understand tree structure and its applications
CLR-5:	utilize hash tables for data storage and use graphs to solve real time problems

Course Outcomes (CO):	At the end of this course, learners will be able to:
CO-1:	devise algorithms to arrange the data in required order and retrieve a specific datum in efficient manner
CO-2:	determine the type of list structure that could be used for solving a problem and implement it using C programming language
CO-3:	devise solutions using linear structures Stack and Queue
CO-4:	express proficiency in usage of tree for solving problems
CO-5:	implement Hash tables for storing data and algorithms to find shortest path between nodes in a graph

Unit-1 - Introduction	Programming in C - Primitive data types, Structures, Self-referential structures, Pointers and structures, Dynamic memory allocation, Matrix multiplication; Data Structure – Definition, Types, ADT, Operations; Mathematical notations - Big O, Omega and Theta, Complexity – Time, Space, Trade off.	15 Hour
Unit-2 - List Structure	Operations on List ADT – Create, Insert, Search, Delete, Display elements; Implementation of List ADT – Array, Cursor based and Linked; Types – Singly, Doubly, Circular; Applications - Sparse Matrix, Polynomial Arithmetic, Joseph Problem	15 Hour
Unit-3 - Stack and Queue	Operations on Stack ADT – Create, Push, Pop, Top; Implementation of Stack ADT – Array and Linked; Applications - Infix to Postfix Conversion, Postfix Evaluation, Balancing symbols, Function Calls, Tower of Hanoi; Operations on Queue ADT - Create, Enqueue and Dequeue; Implementation of Queue ADT – Array and Linked; Types of Queue - Circular, Double ended and Priority Queue, Applications – Scheduling	15 Hour
Unit-4 - Trees and Hashing	Introduction to Trees, Tree traversals, Complete Binary Tree and its height, Binary Search Trees, Need for Balance, Rotation, AVL trees, B Trees, Heaps, trees and array implementations and applications; Hash functions - Introduction, functions, Collision avoidance, Separate chaining, Open Addressing, Linear Probing, Quadratic probing.	15 Hour
Unit-5 - Graph	Introduction to Graph, Graph Traversal, Topological sorting, Minimum spanning tree – Prim's Algorithm, Kruskal's Algorithm, Shortest Path Algorithm - Dijkstra's Algorithm	15 Hour

Lab Experiments	
Lab 1: Implementation of Structures	
Lab 2: Implementation of Structures using Pointers	
Lab 3: Implementation of Matrix Multiplication – Dynamic Memory allocation	
Lab 4: Array Implementation of List	
Lab 5: Implementation of Linked List	
Lab 6: Implementation of Doubly linked List	
Lab 7: Implementation of Stack using array and Linked List	
Lab 8: Implementation of Queue using array and Linked list	
Lab 9: Applications of Stack, Queue	
Lab 10: Implementation of Tree using array	
Lab 11: Implementation of BST using linked list	
Lab 12: Implementation of B-Trees	
Lab 13: Implementation of Graph using Array	
Lab 14: Implementation of Shortest path Algorithm	
Lab 15: Implementation of Minimal Spanning Tree	

Learning Resources	- Seymour Lipschutz, Data Structures with C, McGraw Hill, 2014 - R.F.Gilberg, B.A.Forouzan, Data Structures, 2nd ed., Thomson India, 2005 - A.V.Aho, J.E Hopcroft , J.D.Ullman, Data structures and Algorithms, Pearson Education, 2003 - Mark Allen Weiss, Data Structures and Algorithm Analysis in C, 2nd ed., Pearson Education, 2015 - Reema Thareja, Data Structures Using C, 1st ed., Oxford Higher Education, 2011, - Thomas H Cormen, Charles E Leiserson, Ronald L Revest, Clifford Stein, Introduction to Algorithms 3rd ed., The MIT Press Cambridge, 2014
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Learning Assessment								
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)						Summative Final Examination (40% weightage)
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)				
		Theory	Practice	Theory	Practice	Theory	Practice	
		Theory	Practice	Theory	Practice	Theory	Practice	
Level 1	Remember	25%	-	-	10%	25%	-	
Level 2	Understand	25%	-	-	20%	25%	-	
Level 3	Apply	20%	-	-	30%	20%	-	
Level 4	Analyze	20%	-	-	30%	20%	-	
Level 5	Evaluate	10%	-	-	10%	10%	-	
Level 6	Create	-	-	-	-	-	-	
	Total	100 %		100 %		100 %		

Course Designers		Experts from Higher Technical Institutions		Internal Experts	
Experts from Industry					
1. Dr. Mariappan Vathilingam, Senior Engineering Manager, Uber India Research and Development Pvt Centre, Bangalore.		1. Dr. Venkatesh Raman, Professor, Mathematical Institute of Science		1. Dr. K. Vijaya, SRMIST	
				2. Dr. S. Poornima, SRMIST	
				3. Dr. P. Saranya, SRMIST	